

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High)

Assessment Tool Weightage: 30%

Dr

QUESTIONS		Total Marks: 50	
Q.No	Question	Bloom's Level	Marks
1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this	BL1 – Remember	5

	information about the UPS system.		
2	Course Management System Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.	BL2 – Understand	5
3	University Registrar's Office Management System A university registrar's office maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites; (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; (c) students, including student-id, name, and program; (d) instructors, including identification number, name, department, and title. (e) Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office.	BL2 – Understand	5
4	Hospital Management System Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors.	BL2 – Understand	5

	Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)		
5	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. ■ A department controls a number of PROJECT. Projects have a unique name, number and a single location. ■ Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. ■ Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).	BL2 – Understand	5
6	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember	5
7	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand	5
8	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and	BL2 – Understand	5

	aggregation.		
9	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand	5
10	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember	5

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application

Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q1. UPS Product Tracking Information System (5 marks)

The system tracks shipped items as they move from a retail center of origin, through one or more transportation events, to their final destination.

Entities and Attributes

Entity	Key Attribute(s)	Other Attributes
SHIPPED_ITEM	itemNumber	weight, dimensions, insuranceAmount, destination, finalDeliveryDate
RETAIL_CENTER	uniqueID	type, address
TRANSPORTATION_EVENT	scheduleNumber	type, deliveryRoute

Relationships and Cardinalities

Relationship	Participating Entities & Cardinality	Relationship Attributes
RECEIVED_AT	SHIPPED_ITEM (many) — RETAIL_CENTER (one). Each item is received at exactly one retail center; a retail center receives many items.	-
ROUTED_VIA	SHIPPED_ITEM (many) — TRANSPORTATION_EVENT (many). An item travels via one or more transportation events; a transportation event carries many items.	sequenceNumber (order of the leg in the item's journey)

ER Diagram convention used: rectangles = entity sets, ovals = attributes (underlined = key), diamonds = relationship sets, double rectangles = weak entity sets, double diamonds = identifying relationships. Cardinalities are shown as (min,max) or crow's-foot style 1:1, 1:N, M:N as noted in the table above.

Diagram structure: RETAIL_CENTER —(1,N) RECEIVED_AT (N,1)— SHIPPED_ITEM —(1,N) ROUTED_VIA (M,N)— TRANSPORTATION_EVENT. The M:N relationship ROUTED_VIA is resolved with an associative attribute sequenceNumber to preserve the order of legs in transit.

Q2. Course Management / Exam Scheduling System (5 marks)

Instead of one flat EXAM entity, the schema is normalized into three entity sets connected by relationship sets, with SECTION modeled as a weak entity dependent on COURSE.

Entities and Attributes

Entity	Key Attribute(s)	Other Attributes
COURSE	name (PK)	department, c-number
SECTION (weak entity)	s-number (partial key)	enrollment; identifying owner = COURSE
ROOM	r-number, building (composite PK)	capacity

Relationships and Cardinalities

Relationship	Participating Entities & Cardinality	Relationship Attributes
SECTION_OF (identifying relationship)	COURSE (1) — SECTION (N). Every section belongs to exactly one course; a course has many sections.	-
EXAM_HELD_IN	SECTION (N) — ROOM (1) at a given time. A section's exam is scheduled in one room; a room hosts many exams.	time

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SECTION is a weak entity: it is drawn as a double-outlined rectangle, its discriminator s-number is underlined with a dashed line, and SECTION_OF is drawn as a double-outlined diamond connecting it to its owner COURSE. The attribute 'time' is placed on the EXAM_HELD_IN relationship because a room-time slot is what is actually being scheduled.

Q3. University Registrar's Office Management System (5 marks)

Entities and Attributes

Entity	Key Attribute(s)	Other Attributes
COURSE	course_number	title, credits, syllabus, prerequisites (multivalued)
COURSE_OFFERING (weak entity)	year, semester, section_number (partial key)	timings, classroom; identifying owner = COURSE
STUDENT	student_id	name, program
INSTRUCTOR	instructor_id	name, department, title

Relationships and Cardinalities

Relationship	Participating Entities & Cardinality	Relationship Attributes
OFFERED_AS (identifying relationship)	COURSE (1) — COURSE_OFFERING (N)	-

TEACHES	INSTRUCTOR (1..N) — COURSE_OFFERING (1..N) (an offering may have co-instructors; an instructor teaches many offerings)	-
ENROLLS_IN	STUDENT (M) — COURSE_OFFERING (N)	grade
PREREQUISITE_OF (recursive on COURSE)	COURSE (M) — COURSE (N)	-

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Grade is placed on ENROLLS_IN (not on STUDENT or COURSE_OFFERING) because a grade only makes sense for a specific student-offering pair. Prerequisites are modeled as a recursive many-to-many relationship on COURSE.

Q4. Hospital Management System (5 marks)

ER Diagram description:

Entities and Attributes

Entity	Key Attribute(s)	Other Attributes
PATIENT	SS# (PK)	name, insurance
PHYSICIAN	name (PK)	specialization
TEST (weak entity, log of tests)	test-name, date, time (partial key)	identifying owner = PATIENT

Relationships and Cardinalities

Relationship	Participating Entities & Cardinality	Relationship Attributes
TREATS / DOCTOR_PATIENT	PHYSICIAN (N) — PATIENT (M): a physician treats many patients; a patient may be treated by many physicians	-
UNDERGOES (identifying relationship)	PATIENT (1) — TEST (N): each test-log entry belongs to exactly one patient	-

Corresponding Relational Tables

- Patient (SS#, name, insurance) — SS# is the primary key
- Physician (name, specialization) — name is the primary key
- Test-log (SS#, test-name, date, time) — composite primary key (SS#, test-name, date); SS# is a foreign key referencing Patient

- Doctor-patient (physician-name, SS#) — composite primary key; physician-name references Physician, SS# references Patient
- Patient-history (SS#, test-name, date) — composite primary key; SS# references Patient (this table records which tests appear in a patient's history, distinct from the operational Test-log)

Mapping logic: the weak entity TEST becomes Test-log with the owner's key (SS#) folded in as part of its composite primary key/foreign key. The M:N relationship between PATIENT and PHYSICIAN becomes its own table, Doctor-patient, holding the primary keys of both participating entities.

Q5. Company Project Management System (5 marks)

Entities and Attributes

Entity	Key Attribute(s)	Other Attributes
DEPARTMENT	Dname (PK), Dnumber (alt. key)	Mgr_ssn, Mgr_start_date
DEPT_LOCATIONS (multivalued attr. of DEPARTMENT)	Dnumber, Dlocation (composite PK)	-
PROJECT	Pname (PK), Pnumber (alt. key)	Plocation, controlling Dnumber
EMPLOYEE	Ssn (PK)	Name, Address, Salary, Sex, Bdate, Dno (dept employee belongs to), Super_ssn (supervisor)
DEPENDENT (weak entity)	Dependent_name (partial key)	Birth_date, Relationship; identifying owner = EMPLOYEE

Relationships and Cardinalities

Relationship	Participating Entities & Cardinality	Relationship Attributes
MANAGES	EMPLOYEE (1) — DEPARTMENT (1): one employee manages one department	start_date
CONTROLS	DEPARTMENT (1) — PROJECT (N)	-
WORKS FOR	EMPLOYEE (N) — DEPARTMENT (1)	-
WORKS ON	EMPLOYEE (M) — PROJECT (N)	hours
SUPERVISION	EMPLOYEE (1) — EMPLOYEE (N), recursive: a supervisor oversees many employees	-
DEPENDENTS_OF (identifying relationship)	EMPLOYEE (1) — DEPENDENT (N)	-

ER Diagram convention used: rectangles = entity sets, ovals = attributes (underlined = key), diamonds = relationship sets, double rectangles = weak entity sets, double diamonds =

identifying relationships. Cardinalities are shown as (min,max) or crow's-foot style 1:1, 1:N, M:N as noted in the table above.

This is the classic Elmasri–Navathe COMPANY schema. Dept_locations is drawn as a multivalued attribute (double oval) of DEPARTMENT; alternatively it can be shown as a weak/derived entity if many locations must be queried independently. Hours is an attribute on the M:N relationship WORKS_ON, since it depends on the specific employee-project pairing, not on either entity alone.

Q6. Schema, Instance, and the Three-Schema Architecture (5 marks)

Schema

A schema is the overall design or structure of a database — the description of the database's entities, attributes, relationships, and constraints. It is specified at database design time and, once set, changes only infrequently (schema evolution). Example: a table definition Student(Roll_no, Name, Dept, CGPA) is a schema.

Instance

An instance (also called a database state or snapshot) is the actual data stored in the database at a particular moment in time. It changes every time data is inserted, updated, or deleted. Example: the actual rows currently present in the Student table, e.g. (101, 'Ravi', 'CSE', 8.7), is an instance.

Three Levels of Schema (ANSI/SPARC Architecture)

- Physical (internal) schema — describes how data is actually stored: file organization, indexes, storage structures, access paths. Example: 'Student records are stored as a B+-tree indexed file on Roll_no.'
- Logical (conceptual) schema — describes the structure of the whole database for the community of users: entities, data types, relationships, constraints, hiding physical storage detail. Example: 'Student(Roll_no, Name, Dept, CGPA) with Dept referencing Department(Dept_id).'
- View (external) schema — describes a part of the database relevant to a particular group of users or applications, hiding the rest. Example: an 'Exam Cell' view showing only Roll_no, Name, CGPA and hiding Dept.

Together these three levels provide logical data independence (application programs are unaffected by changes to the conceptual schema) and physical data independence (the conceptual schema is unaffected by changes to the physical storage schema).

Q7. Bank Management System (5 marks)

Entities and Attributes

Entity	Key Attribute(s)	Other Attributes
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BRANCH	branch_code (PK)	branch_name, city, assets
CUSTOMER	customer_id (PK)	name, address, phone
ACCOUNT	account_number (PK)	account_type, balance, open_date
LOAN	loan_number (PK)	amount, loan_type
EMPLOYEE	employee_id (PK)	name, designation, salary

Relationships and Cardinalities

Relationship	Participating Entities & Cardinality	Relationship Attributes
MAINTAINS	BRANCH (1) — ACCOUNT (N)	-
OFFERS	BRANCH (1) — LOAN (N)	-
OWNS/HOLDS	CUSTOMER (M) — ACCOUNT (N): joint accounts are possible	access_date
BORROWS	CUSTOMER (M) — LOAN (N): a loan can have co-borrowers	-
WORKS AT	EMPLOYEE (N) — BRANCH (1)	-
SERVICES (optional)	EMPLOYEE (1) — CUSTOMER (N): a relationship-manager style link	-

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Q8. Extended ER (EER) Model — Generalization, Specialization, Aggregation (5 marks)

The basic ER model captures simple database applications well, but real-world systems often need richer semantics for representing hierarchies among entity types and complex relationships. The EER (Enhanced/Extended ER) model adds three constructs for this.

Generalization

Generalization is a bottom-up process of abstracting a single generic (superclass/superclass) entity type from two or more entity types that share common attributes. Example: CAR and TRUCK, both having license_number and price, can be generalized into a superclass VEHICLE.

Specialization

Specialization is the reverse, top-down process: an entity type (superclass) is divided into subclasses, each of which has attributes or relationships specific to it. Example: EMPLOYEE can be specialized into SECRETARY (typing_speed), ENGINEER (engineer_type), and MANAGER (car_allowance). Specialization/generalization can be:

- Total vs Partial — total means every superclass entity must belong to some subclass; partial means it need not.

- Disjoint vs Overlapping — disjoint means an entity can belong to only one subclass at a time; overlapping means it may belong to several simultaneously.

Aggregation

Aggregation is an abstraction through which a relationship set (together with its participating entity sets) is treated as a higher-level entity set, so that it can itself participate in further relationships. Example: the relationship WORKS_ON between EMPLOYEE and PROJECT can be aggregated into a single higher-level construct so that it can participate in a MONITORS relationship with MANAGER, expressing that a manager monitors the work of an employee on a project, without directly linking MANAGER to the ternary combination.

Notation: generalization/specialization is shown with a circle on the line between superclass and subclasses, labeled 'd' for disjoint or 'o' for overlapping, with a double line if total. Aggregation is shown by enclosing the aggregated relationship and its entities inside a dashed boundary, which is then treated as a single entity for further relationships.

Q9. EER Diagram — Hospital Management System (5 marks)

Extending Q4's hospital model with inheritance and a weak entity:

Entities and Attributes

Entity	Key Attribute(s)	Other Attributes
PERSON (superclass)	person_id (PK)	name, address, phone, DOB
PATIENT (subclass of PERSON)	inherits person_id	SS#, insurance, admission_date
STAFF (subclass of PERSON)	inherits person_id	employee_no, department
PHYSICIAN (subclass of STAFF)	inherits employee_no	specialization, license_no
NURSE (subclass of STAFF)	inherits employee_no	shift
TEST (weak entity)	test-name, date, time (partial key)	result; identifying owner = PATIENT

Relationships and Cardinalities

Relationship	Participating Entities & Cardinality	Relationship Attributes
IS_A (generalization) PERSON -> {PATIENT, STAFF}	Disjoint, Total specialization (o/d, double line)	-
IS_A (generalization) STAFF -> {PHYSICIAN, NURSE}	Overlapping, Partial specialization (a staff member could hold both roles over time; not every staff member is one of these two)	-

TREATS	PHYSICIAN (N) — PATIENT (M)	-
UNDERGOES (identifying relationship)	PATIENT (1) — TEST (N)	-
CARES_FOR	NURSE (N) — PATIENT (M)	-

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PATIENT and STAFF share the common attributes name, address, phone, and DOB, which are factored up into the superclass PERSON — this is generalization. STAFF is further specialized into PHYSICIAN and NURSE based on role-specific attributes. TEST remains a weak entity because a test-log record has no independent existence or unique identifier without knowing which patient it belongs to.

Q10. Role of the Database Administrator (DBA) (5 marks)

The DBA is the person (or team) who has central control over the data and the programs that access that data, and is responsible for the overall management and coordination of the database environment.

Key Responsibilities

- Schema definition — creating and modifying the original database schema by writing DDL statements.
- Storage structure and access-method definition — deciding physical storage organization, indexing, and file structures for performance.
- Schema and physical-organization modification — altering the schema or physical structure to reflect changing organizational needs or improve performance, without disrupting users unnecessarily.
- Granting authorization for data access — deciding which parts of the database different users/roles can access, and what operations (read/write/update) they may perform, enforcing security.
- Routine maintenance — periodic backup of the database, ensuring adequate disk/storage space, monitoring performance and jobs, and applying software upgrades/patches.
- Integrity-constraint specification — defining constraints (keys, referential integrity, domain constraints) to keep the database consistent.
- Recovery and concurrency control — setting up mechanisms to recover from system/media failures and to manage concurrent access without conflicts.
- Liaison with users — interacting with end users and application developers to understand new requirements and resolve data-related issues.

In summary, the DBA safeguards the database's availability, security, integrity, and performance, acting as the bridge between the raw stored data and the many users and applications that depend on it.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____